

Solvency Field Theory V13.2

Expansion as Global Time Production and Boundary-Capacity Growth from Irreversible Writes

V13 Expansion-Time Mechanism Freeze, Native Write Action, Capacity Turnover, and Causal-Front Readout

Expansion is accumulated particle and radiation writing seen as global time-production and boundary-capacity growth.

Frozen Native Master Stack

$$S_{\text{write}} = \int \dot{N}_{\text{write}} dt,$$

$$\dot{N}_{\text{write}} = \sum_a \chi_a \frac{E_a}{h},$$

$$I_{\text{write}} = \eta_I \frac{A_{\text{boundary}}}{A_{\text{write}}},$$

$$H = \frac{\dot{N}_{\text{write}}}{I_{\text{write}}}.$$

Run: SFT_V13_2_R250E_Mechanism_Rollup_Freeze

Branch: V13 Expansion-Time Mechanism

Generated: 2026-06-26

Status: Polished PDF Draft from R250E Freeze

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1 Abstract

SFT V13.2 freezes the expansion/time mechanism of Solvency Field Theory. The central claim is that cosmic expansion is not an added force, field, medium, or dark substance. Expansion is the global boundary-capacity readout of the same irreversible writes that locally produce time and, through compactness gradients, gravity. A locally real particle or radiation mode writes at a current accessible rate set by energy over action, while no-overwrite requires each irreversible write to claim a fresh address. The native expansion equation is

$$H = \frac{\dot{N}_{\text{write}}}{I_{\text{write}}},$$

where $\dot{N}_{\text{write}}$ is current accessible write production and I_{write} is active writable boundary capacity. The same relation admits a causal-front readout $H = c/R$ when R is the settlement radius selected by the same ledger. Matter and radiation differ not by instantaneous write-rate at equal accessible energy, but by write geometry: closed same-cell maintenance dilutes as a^{-3} , while open next-cell propagation dilutes as a^{-4} . Entropy stock, current write flow, and active capacity are separate ledgers; recombination is interpreted as ledger crystallization; and cosmic compactness supplies a recurring $4\pi^2$ boundary-accounting diagnostic. This document freezes the mechanism while explicitly leaving A_{write} derivation, the microscopic Γ/Φ settlement operator, black-hole accessibility, compactness normalization, and observational applications open.

2 Executive Claim and Boundary

V13.2 Expansion-Time Claim

Expansion is accumulated particle and radiation writing seen as global time-production and boundary-capacity growth. Particles and radiation write. No-overwrite requires fresh addresses. Active boundary capacity grows. The fractional growth rate is $H = \dot{N}_{\text{write}}/I_{\text{write}}$, and the same mechanism admits a causal-front readout $H = c/R$.

Immediate Claim Boundary

V13.2 freezes a mechanism. It does not derive A_{write} , does not solve the microscopic Γ/Φ settlement operator, does not derive the black-hole accessibility factor f_{access} , does not normalize C_A at horizons, and does not claim DESI, CMB, BAO, H_0 , structure-growth, or V5 black-hole entropy application fits.

3 V13 Continuity Map

V13.2 is the expansion/time branch of the V13 program. It is not a standalone cosmology bolt-on. It is the global readout of the same locally-real write process.

V13 One-Write Continuity

One irreversible write



V13.0 Particle

local closure/tick
time production

V13.1 Gravity

local settlement
compactness
gradient

V13.2 Expansion

global capacity
boundary growth

The frozen continuity sentence is:

**Time production, gravity, and expansion are not three separate costs.
They are three readouts of the same write process.**

4 Native Ontology

SFT V13.2 does not begin from an imported Einstein-Hilbert action, a background medium, a lattice, an aether, a foam, or a new particle sector. The native objects are:

Object	Native role
Irreversible write	A completed physical update into permanent history.
Locally real support	The real particle/radiation process that writes.
Boundary capacity	The active writable-address budget required by no-overwrite.
Compactness	The local intensity of energy/settlement demand.
No-overwrite	The rule that completed history cannot be written again.

A write is an irreversible update of realized history. Once written, history is locked. Only the next write is editable. Therefore:

ongoing writes $\Rightarrow$ ongoing fresh writable capacity.

That is the mechanism-level origin of expansion in V13.2.

5 Native First, Translation Second

V13.2 follows a strict presentation rule:

Equation Discipline

Native SFT equations come first. GR/cosmology forms appear directly underneath only as labeled translations or correspondences. The GR-readable form is not the native action.

The native write stack is:

$$S_{\text{write}} = \int \dot{N}_{\text{write}} dt, \quad (1)$$

$$\dot{N}_{\text{write}} = \sum_a \chi_a \frac{E_a}{h}, \quad (2)$$

$$I_{\text{write}} = \eta_I \frac{A_{\text{boundary}}}{A_{\text{write}}}, \quad (3)$$

$$H = \frac{\dot{N}_{\text{write}}}{I_{\text{write}}}. \quad (4)$$

Here E_a is accessible active energy in channel a , h is Planck's constant, χ_a is an accessibility/channel factor, A_{write} is the boundary-area quantum carried from V13.1, and η_I is the boundary capacity convention. For ordinary accessible non-horizon matter and radiation at the coarse-grained level, $\chi_a \simeq 1$. Horizon-sequestered energy requires an accessibility factor and is not assumed to contribute as full exterior-accessible expansion flow.

6 No-Overwrite and Capacity Exhaustion

Under no-overwrite, a fixed finite writable capacity cannot support indefinite writes. If remaining capacity is $I_{\text{remaining}}$, then

$$t_{\text{exhaust}} = \frac{I_{\text{remaining}}}{\dot{N}_{\text{write}}}.$$

For the whole active ledger governed by Eq. (4),

$$t_{\text{exhaust}} = \frac{1}{H}.$$

This is the break-even result: the Hubble time is the active capacity-turnover time.

Capacity Claim Boundary

Do not state this as “expansion instantly causes time.” The safe statement is that capacity growth is necessary for indefinite time production under no-overwrite.

7 A_{write} Capacity Form and Friedmann Translation

Using the quarter-boundary convention

$$\eta_I = \frac{1}{4},$$

the R242E capacity derivation gives the native energy-density form:

$$H^2 = \frac{2A_{\text{write}}c\rho_E}{3\pi\hbar}.$$

With

$$G_A = \frac{A_{\text{write}}c^3}{\hbar}, \quad \rho_E = \rho_m c^2,$$

the high-value GR-readable translation is

$$H^2 = \frac{2G_A \rho_m}{3\pi}.$$

This is the SFT write/capacity relation expressed in mass-density language. It is not standard flat GR Friedmann equivalence.

8 Causal-Front Readout: $H = c/R$

The same expansion rate also has a geometric readout. If the settlement front advances causally,

$$\frac{dR}{dt} = c,$$

then the fractional front-growth rate is

$$H = \frac{c}{R}.$$

This is equivalent to $H = \dot{N}_{\text{write}}/I_{\text{write}}$ only when R is the causal settlement radius selected by the same ledger. Equating the two forms gives

$$\frac{\dot{N}_{\text{write}}}{I_{\text{write}}} = \frac{c}{R}.$$

Using

$$\dot{N}_{\text{write}} = \frac{E_{\text{total}}}{h}, \quad I_{\text{write}} = \eta_I \frac{4\pi R^2}{A_{\text{write}}},$$

selects

$$R = \frac{E_{\text{total}} A_{\text{write}}}{4\pi \eta_I h c}.$$

For $E_{\text{total}} = Mc^2$ and $A_{\text{write}} = G_A \hbar / c^3$,

$$R = \frac{G_A M}{8\pi^2 \eta_I c^2}.$$

With $\eta_I = 1/4$,

$$R = \frac{r_s}{4\pi^2}, \quad r_s = \frac{2G_A M}{c^2}.$$

R246E Preservation

$H = c/R$ is the causal-front readout of the same capacity-turnover rate, provided R is the selected settlement radius. It is not a statement about arbitrary radius.

9 Matter and Radiation as Write Geometry

V13.2 separates instantaneous write rate from evolutionary geometry. At one instant,

$$\dot{N} = \frac{E_{\text{accessible}}}{h}.$$

Matter and radiation differ by write geometry.

Channel	Native geometry	Scaling	Meaning
Matter	closed same-cell maintenance	$\rho_m \propto a^{-3}$	comoving rest-energy write demand persists while volume grows
Radiation	open next-cell propagation	$\rho_r \propto a^{-4}$	number density dilutes and each photon energy redshifts as wavelength stretches

The extra a^{-1} in radiation is not a second instantaneous write-rate bonus. It is the evolution of open next-cell propagation through expanding capacity.

10 Stock, Flow, Capacity, and Accessibility

V13.2 freezes the ledger separation:

$$\text{entropy stock} \neq \text{current write flow} \neq \text{active capacity}.$$

Entropy stock is already-written locked history. Write flow is current irreversible write production. Capacity is the active writable-address budget. The expansion equation uses current flow in the numerator, not entropy stock.

10.1 Black-Hole Guardrail

Black holes require a split ledger:

Ledger	Black-hole handling
Exterior gravity	sees full M
Internal write demand	may carry Mc^2/h as sequestered demand
Expansion-accessible flow	$\dot{N}_{\text{BH,exp}} = f_{\text{access}} Mc^2/h$
Entropy stock	horizon/interior stock, blocked from numerator

The amendment to the process-agnostic energy null is:

$$\text{equal accessible active energy gives equal instantaneous expansion write-rate.}$$

The accessibility law f_{access} is open.

11 Recombination as Ledger Crystallization

Recombination is ledger crystallization. Before recombination:

$$\text{ionized photon-baryon plasma} = \text{coupled ledger fluid}.$$

During recombination and photon decoupling:

charged carriers close into neutral matter structures, opacity drops, and last-scattering history locks

After recombination:

matter = same-cell neutral maintenance, radiation = free next-cell photon propagation.

The CMB must be separated into two roles:

CMB pattern	locked historical stock
CMB photons	current radiation flow

Optical depth is retained as a standard translation:

$$\tau \gg 1 \Rightarrow \text{coupled ledger}, \quad \tau \sim 1 \Rightarrow \text{last scattering}, \quad \tau \ll 1 \Rightarrow \text{free streaming}.$$

The last-scattering surface is a historical surface, not automatically the current active settlement boundary.

12 Cosmic Compactness Diagnostic

V13.1 defined compactness as

$$C_A(E, R) = \frac{A_{\text{write}} E}{\hbar c R}.$$

Using $G_A = A_{\text{write}} c^3 / \hbar$ and $E = M c^2$,

$$C_A = \frac{G_A M}{c^2 R}.$$

For a homogeneous sphere with $R = c/H$, if

$$H^2 = \alpha G_A \rho_m,$$

then

$$C_A = \frac{4\pi}{3\alpha}.$$

Standard flat GR critical density uses $\alpha = 8\pi/3$, so

$$C_A = \frac{1}{2}.$$

SFT R242E uses $\alpha = 2/(3\pi)$, so

$$C_A = 2\pi^2.$$

Therefore

$$\frac{C_{A,\text{SFT}}}{C_{A,\text{GR}}} = 4\pi^2.$$

The half-compactness intuition survives only as the boundary-normalized SFT statement:

$$\frac{C_A}{4\pi^2} = \frac{1}{2}.$$

The current C_A convention gives $C_A = 1/2$ at the standard Schwarzschild horizon, so $C_A = 1$ horizon language requires a separate normalization gate. In GR translation, the safer horizon variable is $u = r_s/r = 2C_A$.

13 Microscopic Settlement Operator: Open but Located

V13.2 does not derive the microscopic settlement relaxation operator. The open bridge is

$$\Delta \ln S_i = L_{\text{eff}} u_i - \Phi_{\text{relax}}(A_{\text{write}}; Q, \Lambda, \tau, D).$$

This pulls forward three older structures:

- R192M: per-cycle logarithmic settlement gain;
- V12.1: settlement-frontier / settlement-volume coordinate;
- V11.1B: finite-memory lag kernel.

If Φ_{relax} is common across modes, it cancels from ratios:

$$\ln \left(\frac{S_i}{S_j} \right) = L_{\text{eff}}(u_i - u_j).$$

However, Φ_{relax} and Γ_{settle} are not derived in V13.2. They are deferred to a microscopic settlement-operator program.

14 Promoted, Open, and Quarantined

Category	Item	Freeze handling
Promoted	Expansion as capacity growth	Native mechanism: $H = \dot{N}_{\text{write}}/I_{\text{write}}$.
Promoted	Same write, three readouts	Local time, local settlement/gravity, global capacity demand.
Promoted	No-overwrite capacity requirement	Fixed capacity exhausts after $I/\dot{N}$; whole-ledger turnover is $1/H$.
Promoted	Matter/radiation geometry	Same-cell matter and next-cell radiation evolution frozen.
Promoted	Stock/flow/capacity split	Locked stock, current flow, and active capacity are distinct ledgers.
Promoted	Recombination	Ledger crystallization; CMB pattern stock vs CMB photon flow.
Promoted	$4\pi^2$ diagnostic	R242/R246/R249 boundary-accounting recurrence.
Open	A_{write} derivation	Used consistently from V13.1; not derived.
Open	Γ/Φ operator	Operator family located; not solved.
Open	Black-hole f_{access}	Split ledger selected; accessibility law open.
Open	C_A normalization	Horizon translation requires normalization gate.
Quarantined	DESI/CMB/BAO/ H_0 applications	Downstream application tracks only.
Quarantined	V5 factor-14 repro	Requires stock/flow black-hole entropy reproduction.

15 Gate Rollup

Gate	Title	Status	Frozen result
R240E	Source inventory	PASS	Old expansion/cosmology claims cataloged; applications quarantined.
R241E	Master equation	PASS	$H = \dot{N}_{\text{write}}/I_{\text{write}}$ translated into particle-time language.

Gate	Title	Status	Frozen result
R241.5E	Local/global write audit	PASS	One write has local time, local gravity, and global capacity readouts.
R241.75E	Native action / GR translation	PASS	Native write action is primary; GR forms are translations.
R242E	A_{write} Friedmann form	PASS	$H^2 = 2A_{\text{write}}c\rho_E/(3\pi\hbar)$ with $\eta_I = 1/4$.
R243E	Matter/radiation geometry	PASS	Closed matter a^{-3} ; open radiation a^{-4} .
R244E	Energy null	PASS	Equal accessible active energy gives equal instantaneous write rate.
R245E	Capacity/time audit	PASS	$t_{\text{exhaust}} = 1/H$ for whole active ledger.
R245B/C	Settlement bridge	PARTIAL/PASS	Naive Γ blocked; log-gain Φ_{relax} operator family carried open.
R246E	$H = c/R$ bridge	PASS	Capacity turnover equals causal-front readout for selected R .
R247E/B	Stock/flow/accessibility	PASS/PARTIAL	Stock, flow, capacity, and horizon accessibility separated.
R248E	Recombination	PASS	Ledger crystallization and CMB stock/flow split.
R249E	Compactness	PASS	SFT/GR compactness ratio $4\pi^2$.
R250E	Mechanism freeze	PASS	V13.2 mechanism frozen.

16 Final V13.2 Statement

Frozen Statement

V13.2 establishes expansion as the global capacity-growth readout of irreversible writing. Particles and radiation write. No-overwrite requires fresh addresses. Active boundary capacity grows. The fractional growth rate is $H = \dot{N}_{\text{write}}/I_{\text{write}}$. The same mechanism admits a causal-front readout $H = c/R$. Time production, gravity, and expansion are not three separate costs. They are three readouts of the same write process: local time as the tick count, local gravity as compactness-gradient settlement, and global expansion as boundary-capacity turnover.

Acknowledgments

The V13.2 expansion-time mechanism was developed through the SFT collaboration with Shaun Fosmark, ChatGPT/5.5, and parallel side-review from Opus. The document preserves the R240E–R250E mechanism chain and quarantines observational applications to later companion work.

A Equation Index

ID	Equation	Meaning
write action	$S_{\text{write}} = \int \dot{N}_{\text{write}} dt$	Native action counts irreversible writes.
write flow	$\dot{N}_{\text{write}} = \sum_a \chi_a E_a / \hbar$	Current accessible active write production.
capacity	$I_{\text{write}} = \eta_I A_{\text{boundary}} / A_{\text{write}}$	Active writable boundary capacity.
expansion	$H = \dot{N}_{\text{write}} / I_{\text{write}}$	Capacity-turnover rate.
exhaustion	$t_{\text{exhaust}} = I_{\text{remaining}} / \dot{N}_{\text{write}}$	Frozen-capacity exhaustion time.
Friedmann translation	$H^2 = 2G_A \rho_m / (3\pi)$	GR-readable translation of R242E.
causal front	$H = c/R$	Geometric readout for selected settlement radius.
selected radius	$R = E_{\text{total}} A_{\text{write}} / (4\pi \eta_I \hbar c)$	Radius chosen by write/capacity ledger.
compactness	$C_A = A_{\text{write}} E / (\hbar c R)$	V13.1/V13.2 compactness diagnostic.

ID	Equation	Meaning
settlement operator	$\Delta \ln S_i = L_{\text{eff}} u_i - \Phi_{\text{relax}}$	Open microscopic bridge.

B Application Roadmap

- **V13.2A:** DESI / late-time crossing application.
- **V13.2B:** CMB / recombination compatibility and sound-horizon interface.
- **V13.2C:** black-hole entropy / V5 factor-14 stock-flow repro.
- **V13.2D:** BAO / H_0 / structure-growth application.
- **V13.3 or companion:** microscopic Γ/Φ settlement operator.
- **V13.x guardrail:** compactness normalization and black-hole horizon convention.